MATHEMATICS FORMULA SHEET (CLASS X)

Sindh Textbook Board Karachi — Algebraic, Trigonometric & Geometric Formulae

1. Algebraic Identities & Quadratic Equations

- **Quadratic Formula:** For $ax^2 + bx + c = 0$, $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$.
- **Discriminant:** $D = b^2 - 4ac$ (Determines nature of roots).
- **Cubic Expansion:** $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.
- **Difference of Cubes:** $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$.

2. Trigonometry Identities

Fundamental Identity	Derived Form 1	Derived Form 2
$\sin^2\theta + \cos^2\theta = 1$	$\sin^2\theta = 1 - \cos^2\theta$	$\cos^2\theta = 1 - \sin^2\theta$
$1 + \tan^2\theta = \sec^2\theta$	$\tan^2\theta = \sec^2\theta - 1$	$\sec^2\theta - \tan^2\theta = 1$
$1 + \cot^2\theta = \operatorname{cosec}^2\theta$	$\cot^2\theta = \operatorname{cosec}^2\theta - 1$	$\operatorname{cosec}^2\theta - \cot^2\theta = 1$

3. Coordinate Geometry & Matrices

- **Distance Formula:** $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.
- **Mid-Point Formula:** $M = ((x_1 + x_2)/2 , (y_1 + y_2)/2)$.
- **Determinant of Matrix A = [[a, b], [c, d]]:** $|A| = ad - bc$.

4. Practical Geometry & Mensuration

Shape	Area Formula	Perimeter / Volume
Circle	$Area = \pi r^2$	Circumference = $2 \pi r$
Cylinder	Surface = $2 \pi r h$	Volume = $\pi r^2 h$
Sphere	Surface = $4 \pi r^2$	Volume = $(4/3) \pi r^3$